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Programming in Python

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Programming in Python

1. Write a program in Python to Compute Sum, Subtraction, Multiplication, Division, and Exponent of variables input by the user.

```
A = input("Enter first number:
");
B = input("Enter Second
number: ");

C = int(A) + int(B);
D = int(A) - int(B);
E = int(A) * int(B);
F = int(A) / int(B);
G = int(A)**int(B);

print("The sum is: ",C);
print("The Subtraction is: ",D);
print("The multiplication is: ",E);
print("The division is: ",F);
print("The exponent is: ",G);
```

Output:

```
Enter first number: 8
Enter Second number: 2
The sum is: 10
The Subtraction is: 6
The multiplication is: 16
The division is: 4.0
The exponent is: 64
```

2. Compute area of following shapes: circle, rectangle, triangle, square, trapezoid and parallelogram.

Circle

```
int(input("Enter radius: "))
pi = 3.14 r =
area = pi*(r*r)
print("Area of circle is: ",area)
```

Output:

```
Enter radius: 4
Area of circle is: 50.24
```

Triangle

```
h = int(input("Enter Height: "))
b = int(input("Enter base: "))
area = 1/2*b*h
print("Area of triangle is:",area)
```

Output:

```
Enter Height: 10
Enter base: 7
Area of triangle is: 35.0
```

Programming in Python

Rectangle

```
l = int(input("Enter length: "))
b = int(input("Enter breadth: "))
area = l*b
print("Area of reactangle is:",area)
```

Output:

```
Enter length: 8
Enter breadth: 12
Area of reactangle is: 96
```

Trapezoid

```
a = int(input("Enter upper base length: "))
b = int(input("Enter lower base length: "))
h = int(input("Enter height: "))
area=1/2*(a*b)*h
print("Area of trapezoid is: ",area)
```

Output:

```
Enter upper base length: 7
Enter lower base length: 11
Enter height: 9
Area of trapezoid is: 346.5
```

Parallelogram

```
h = int(input("Enter Height: "))  
b = int(input("Enter base: "))  
area = b*h  
print("Area of parallelogram is:",area)
```

Output:

```
Enter Height: 11  
Enter base: 7  
Area of parallelogram is: 77
```

Square

```
s = int(input("Enter side: "))  
area = s*s  
print("Area of square is:",area)
```

Output:

```
Enter side: 6  
Area of square is: 36
```

Programming in Python

3. Write a program in Python to Compute the Volume of following shades:- Cylinder, Cube, Cone, Sphere.

Cylinder

```
pi = 3.14 r = int(input("Enter radius:"))  
h = int(input("Enter height: "))  
volume = pi*(r*r)*h  
print("volume of cylinder is: ",volume)
```

Output:

```
Enter radius: 4  
Enter height: 12  
volume of cylinder is: 602.88
```

Cube

```
s = int(input("Enter side: "))  
volume = s*s*s  
print("volume of cube is:",volume)
```

Output:

```
Enter side: 5  
volume of cube is: 125
```


Sphere

```
pi = 3.14 r = int(input("Enter radius: "))  
h = int(input("Enter height: "))  
volume = 1/3*pi*(r*r)*h  
print("volume of sphere is:",volume)
```

Output:

```
Enter radius: 4  
Enter height: 8  
volume of sphere is: 133.97333333333333
```

Cone

```
pi = 3.14 r = int(input("Enter radius: "))  
volume = 4/3*pi*(r*r*r)  
print("volume of cone is:",volume)
```

Output:

```
Enter radius: 5  
volume of cone is: 523.3333333333334
```

Programming in Python

4. Write a program in Python to Compute and print roots of Quadratic equations where the values of a, b, and c entered by user.

```
A = int(input("Enter First number: "))
B = int(input("Enter Second number: "))
C = int(input("Enter Third number: "))

x = ((B) + (B**2 - (4*A*C))**0.5)/2*A
y = ((B) - (B**2 - (4*A*C))**0.5)/2*A

print(x)
print(y)
```

Output:

```
Enter First number: 12
Enter Second number: 29
Enter Third number: 15
240.0
108.0
```

Programming in Python

Q.5 Write a program in Python to print numbers up to Which are not Divisible by 3, 6, 9.

```
A = int (input("Enter any number: "))
print("The numbers from 1 to",A)
print("not divisible by 3, 6, 9 are: ")

for i in range(1,A+1):
    if i%3 != 0:
        print(i)
```

Output:

```
Enter any number: 20
The numbers from 1 to 20
not divisible by 3, 6, 9 are:
1
2
4
5
7
8
10
11
13
14
16
17
19
20
```

Programming in Python

Q.6 Write a program in Python to determine whether a triangle is isosceles or not.

```
A = input("Enter first value: ")
B = input("Enter second value: ")
C = input("Enter third value: ")

if A == B or A == C or B == C:
    print("it is isosceles triangle ")

else:
    print("it is not an isosceles triangle")
```

Output:

```
Enter first value: 6
Enter second value: 7
Enter third value: 6
yes it is isosceles triangle
```

Programming in Python

Q.7 Write a program in Python to print multiplication table of a number input by user.

```
n = int(input("Enter the number to print the tables for: "))
```

```
for i in range(1,11):  
    print(n,"x",i,"=",n*i)
```

Output:

```
Enter the number to print the tables for: 8  
8 x 1 = 8  
8 x 2 = 16  
8 x 3 = 24  
8 x 4 = 32  
8 x 5 = 40  
8 x 6 = 48  
8 x 7 = 56  
8 x 8 = 64  
8 x 9 = 72  
8 x 10 = 80
```

Programming in Python

Q.8 Write a program in Python to print sum of natural number input by user.

```
A = int(input("Enter the number: "))

if(A<0):
    print("Enter any positive number")
else:
    sum = 0
    while(A>0):
        sum = sum + A
        A = A-1
    print("Sum till entered number is: ",sum)
```

Output:

```
Enter the number: 8
Sum till entered number is: 36
```

Programming in Python

Q9. Write a program in Python to print the Fibonacci series up to n numbers input by user.

```
A = int(input("Enter the number: "))
n1, n2 = 0, 1

print("Fibonacci Series: ", n1, n2, end="")
for i in range(2, A):
    n3 = n1 + n2
    n1 = n2
    n2 = n3
    print(n3, end=" ")
```

Output:

```
Enter the number: 12
Fibonacci Series:  0 1 1 2 3 5 8 13 21 34 55 89
```

Programming in Python

Q10. Write a program in Python to compute the factorial of a Given Number input by user.

```
A = int(input("Enter the number: "))
x = 1
for i in
range(1,A+1):
    x = x*i

print ("The factorial of given number is: ",x," ")
```

Output:

```
Enter the number: 8
The factorial of given number is:  40320
```


Programming in Python

Q11. Write a program in Python to count the occurrence of digit 5 in a given integer number input by the user.

```
A = int(input("Enter number:"))  
B = int(input("Digit to find the occurrence: "))
```

```
count=0  
while(A>0):  
    rem=A%10  
  
    if rem==5:  
        count=count+1  
  
    A=A//10  
  
print("occurrence is: ",count)
```

Output:

```
Enter number:455535  
Digit to find the occurrence: 5  
occurrence is: 4
```

Programming in Python

Q.12 Write a program in Python to print Geometric and Harmonic means of a series input by the user.

```
import math
def compute(a,b):
    AM=(a+b)/2
    GM=math.sqrt(a*b)
    HM=(GM*GM)/AM
    return HM

def geo(c,d):
    GM=math.sqrt(c*d)
    return GM

a=int(input("Enter the value of a in harmonic mean: "))
b=int(input("Enter the value of b in harmonic mean: "))
c=int(input("Enter the value of a in geometric mean: "))
d=int(input("Enter the value of b in geometric mean: "))

HM=compute(a,b)
GM=geo(c,d)

print("The Harmonic mean is: ", HM)
print("The Geometric mean is: ", GM)
```

Output:

```
Enter the value of a in harmonic mean: 3
Enter the value of b in harmonic mean: 12
Enter the value of a in geometric mean: 5
Enter the value of b in geometric mean: 11
The Harmonic mean is:  4.8
The Geometric mean is:  7.416198487095663
```

Programming in Python

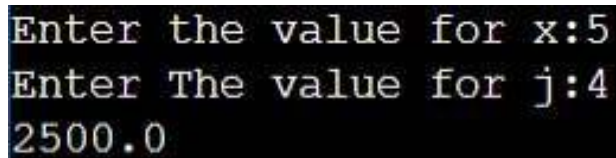
Q.13 Write a program in python to Evaluate the following expressions

a. $x - x^2/2! + x^3/3! - x^4/4! + \dots x^n/n!$

b. $x - x^3/3! + x^5/5! - x^7/7! + \dots x^n/n!$

```
x = int(input("Enter the value for x: "))
j = int(input("Enter The value for j: "))
Sum = 0
n = 1
m = 1
for i in range(1,x+1):
    for i in range(1,j+1):
        n = n*i
    Sum += (m*(x**i)/n*j)
print(Sum)
```

Output:

A screenshot of a terminal window with a black background and white text. It shows the execution of the Python program. The first prompt is "Enter the value for x:" followed by the input "5". The second prompt is "Enter The value for j:" followed by the input "4". The final output is "2500.0".

```
Enter the value for x:5
Enter The value for j:4
2500.0
```

Programming in Python

Q.14 Write a program in Python to print all possible combinations of 4, 5 and 6

```
A = int(input("Enter First number: "))
B = int(input("Enter Second number: "))
C = int(input("Enter Third number: "))
print("Combinations of these numbers are:")

my_list = []

my_list.append(A)
my_list.append(B)
my_list.append(C)

for i in range(0,3):
    for j in range(0,3):
        for k in range(0,3):
            if(i!=j & j!=k & k!=i):
                print(my_list[i], my_list[j], my_list[k])
```

Output:

```
Enter First number: 4
Enter Second number: 5
Enter Third number: 6
Combinations of these numbers are:
4 5 6
4 6 5
5 4 6
5 6 4
6 4 5
6 5 4
```

Programming in Python

Q.15 Write a program in Python to print prime number within a specific range

```
A = int(input("Enter initial number: "))
B = int(input("Enter number till you want to print: "))

for i in range(A,B):
    if i>1:
        for j in range(2,i):
            if(i%j)==0:
                break
        else:
            print(i,end=" ")
```

Output:

```
Enter initial number: 2
Enter number till you want to print: 20
2 3 5 7 11 13 17 19
```

Programming in Python

Q.16 Write a program in Python to print the count of ages of persons above 60 and below 90.

```
n = int(input("Number of persons : "))
print("Enter their ages : ")

Ages = []
for i in range(n):

    Ages.append(int(input()))

count = 0
for x in Ages:
    if (x>=60 and x<=90):
        count = count + 1

print("Number of person above 60 and below 90 are : ", count)
```

Output:

```
Number of persons : 3
Enter their ages :
67
75
92
Number of person above 60 and below 90 are : 2
```

Programming in Python

Q.17 Write a program in Python to compute the transpose of a matrix.

```
x=[[4,2], [3,4],[5,8]]  
y=[[0,0,0],[0,0,0]]  
for i in range(len(x)):  
    for j in range(len(x[0])):  
        y[j][i]=x[i][j]  
for a in y:  
    print(a)
```

Output:

```
[4, 3, 5]  
[2, 4, 8]
```

Programming in Python

Q.18 Write a program in Python to perform the following operations on two matrices

a) Multiplication

b) Addition

c) Subtraction

a) Multiplication

```
x=[[5,4,0],
   [4,8,2],  [9,7,3]]

y=[[3,9,4],
   [5,16,1], [2,20,9]]

z=[[0,0,0], [0,0,0],
   [0,0,0]]
for i in range(len(x)):
    for j in
        range(len(x[0])):
            z[i][j]=x[i][j]*y[j][i]

print(z)
```

Output:

```
[15, 20, 0]
[36, 128, 40]
[36, 7, 27]
```


Programming in Python

b) Addition

```
x=[[4,2],[3,4],  
  [5,8],[9,0]]  
  
y= [[0,7],[0,0],  
    [5,7],[4,9]]  
  
z= [[0,0],[0,0],  
    [0,0],[0,0]]  
for i in range(len(x)):  
    for j in range(len(x[0])):  
        z[i][j]=x[i][j]+y[i][j]  
    print(z)
```

Output:

```
[4, 9]  
[3, 4]  
[10, 15]  
[13, 9]
```

c) Subtraction

```
x=[[4,2],[3,4],  
  [5,8],  
  [9,0]]  
y=[[0,7],  
  [0,0],  
  [5,7],  
  [4,9]]  
  
Z = [[0,0],  
      [0,0],  
      [0,0],[0,0]]  
for i in range(len(x)):  
    for j in  
        range(len(x[0])):  
        z[i][j]=x[i][j]-y[i][j]  
    print(z)
```

Programming in Python

Output:

```
[4, -5]  
[3, 4]  
[0, 1]  
[5, -9]
```

Programming in Python

Q.19 Write a program in Python to count the occurrence of vowels.

```
def Check_Vow(string, vowels):  
    string = string.casefold()  
    count = {}.fromkeys(vowels, 0)  
    for character in string:  
        if character in count:  
            count[character] += 1  
    return count  
vowels = 'aeiou'  
string = "Hey,what's going on"  
print (Check_Vow(string, vowels))
```

Output:

```
{'a': 1, 'e': 1, 'i': 1, 'o': 2, 'u': 0}
```

Programming in Python

Q20. Write a program in Python to count the total no. of vowels in a word.

```
string=input("Enter string:")
vowels=0
for i in string:

    if(i=='a' or i=='e' or i=='i' or i=='o' or i=='u' or i=='A' or i=='E' or i=='I' or i=='O' or
i=='U'):
        vowels=vowels+1
print("Number of vowels are:")
print(vowels)
```

Output:

```
Enter string:What is the main reason behind that
Number of vowels are:
11
```

Programming in Python

Q21. Write a program in Python to print Determine whether a string is palindrome or not.

```
a="vikash"  
reverse = a[::-1]  
if a == reverse:  
    print("string is palindrome")  
else:  
    print("string is not a palindrome")
```

Output:

```
string is palindrome
```

Programming in Python

Q22. Write a program in Python to perform the following operations on list of number.

- a) Insert an element**
- b) Delete an element**
- c) Sort the list**
- d) Delete the list**

a) To insert an element

```
l=[50,70,20,80,56,43]
l.insert(4,'R')
print(l)
```

Output:

```
[50, 70, 20, 80, 'R', 56, 43]
```

b) To delete an element

```
l=[50,70,20,80,56,43]
l.pop(2)
print(l)
```

Output:

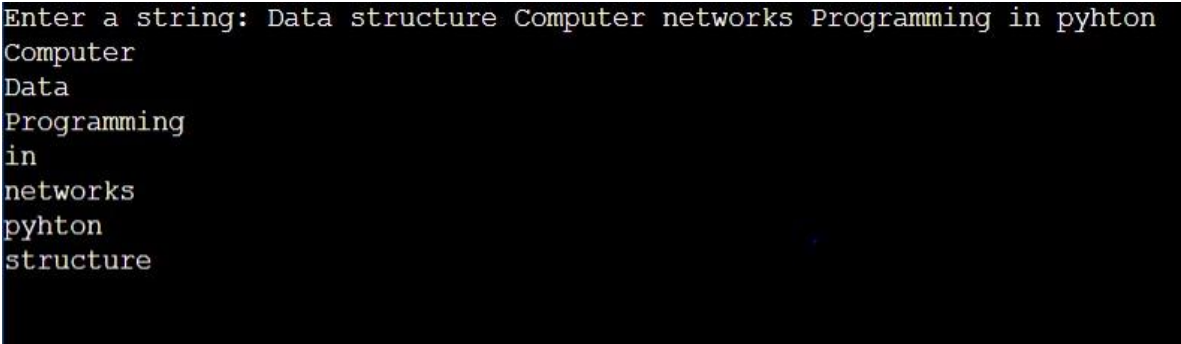
```
[50, 70, 80, 56, 43]
```

Programming in Python

23. Write a program in Python to display the word after sorting in alphabetical order.

```
my_str = input("Enter a string:")  
words = my_str.split()  
words.sort()  
for word in words:  
    print(word)
```

Output:



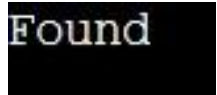
```
Enter a string: Data structure Computer networks Programming in pyhton  
Computer  
Data  
Programming  
in  
networks  
pyhton  
structure
```

Programming in Python

24. Write a program in Python to perform sequential search on a list of given number.

```
def search(List, n):  
    for i in range(len(List)):  
        if List[i] == n:  
            return True  
    return False  
  
List = [1, 2, 'harry', 4, 'geeks', 6]  
n = 'geeks'  
  
if search(List, n):  
    print("Found")  
else:  
    print("Not Found")
```

Output:



Found

Programming in Python

Q25. Write a program in Python to perform sequential search on ordered list of given number.

```
l=[1,20,89,75,65,43,10]
    l.sort()
print(l)
count=0
n=int(input("Enter the number you want to search: "))
    for i in l:
if (i==n):
count=1
break
    if (count==1):
        print("Number is present in the list.")
else:
        print("Number is not present in the list.")
```

Output:

```
[1, 10, 20, 43, 65, 75, 89]
Enter the number you want to search: 43
Number is present in the list.
```

Programming in Python

Q26. Write a program in Python to maintain practical notebook as per their serial number in library using python dictionary.

```
d1 = {}  
d1[5] = "print number upto n 3,2,4,5"  
d1[1] = "compute sum,multiplication"  
d1[4] = " compute value of square"  
d1[3] = " compute volume of following 3 shapes"  
d1[2] = " compute area of following shapes"  
d = {}  
d =sorted(d1.items())  
for i,j in d:  
    print("program number : ", i, ":", j)
```

Output:

```
program number : 1 : compute sum,multiplication  
program number : 2 :  compute area of following shapes  
program number : 3 :  compute volume of following 3 shapes  
program number : 4 :  compute value of square  
program number : 5 : print number upto n 3,2,4,5
```

Programming in Python

Q28. Write a program in Python to check whether a no. is in a given range using function.

```
def FUN(st,end,key):  
    A=set()  
    for i in range (st,end+1):  
        A.add(i)  
    print("SET A is: ",A)  
    if key in A:  
        print(key,"is present in A")  
    else:  
        print(key,"is absent in A")  
  
    a=int(input("Enter the starting point of range: "))  
    b=int(input("Enter the end point of range: "))  
    n=int(input("Enter the element to find: "))  
    if  
a>=b:  
    print("Kindly enter first value as smaller: ")  
else:  
    FUN(a,b,n)
```

Output:

```
Enter the starting point of range: 3  
Enter the end point of range: 10  
Enter the element to find: 7  
SET A is: {3, 4, 5, 6, 7, 8, 9, 10}  
7 is present in A
```

Programming in Python

Q29. Write a program in Python function that accepts a string and calculate number of upper case letters and lower case letters available in a string.

```
def
check(string):
    upper = 0
    lower = 0
    for x in string:
        if(x.isupper() == True):
            upper = upper + 1
        if(x.islower() == True):
            lower = lower + 1
    print("Uppercase letter =", upper)
    print("Lowercase letter =", lower)

y = str(input("Enter any string: "))
check(y)
```

Output:

```
Enter any string: Who Discovered Python language
Uppercase letter = 3
Lowercase letter = 24
```

Programming in Python

Q30. Write a program in Python to find a max of three numbers using functions.

```
def max(x, y, z):  
    if (x>y and x>z):  
        return x  
    elif (y>z and y>x):  
        return y  
    else:  
        return z  
  
a = int(input(" Enter First number: "))  
b = int(input(" Enter Second number: "))  
c = int(input(" Enter Third number: "))  
  
result = max(a,b,c)  
print(" Maximum Number is :", result)
```

Output:

```
Enter First number: 75  
Enter Second number: 63  
Enter Third number: 85  
Maximum Number is : 85
```

Programming in Python

Q31. Write a program in Python to multiply all the numbers using lists using functions.

```
def
multiply_ele(a):
    n=1
    for i in a:
        n = n*i
    return n
a = [10, 11, 12, 13, 14]
print("The multiplication of all the elements of a is: ",multiply_ele(a))
```

Output:

```
The multiplication of all the elements of a is: 240240
```

Programming in Python

Q32. Write a program in Python to solve the Fibonacci sequence using recursion.

```
def recur(n):  
    if n <= 1:  
        return n  
    else:  
        return((n-1) + (n-2))  
  
x = 10  
if x <= 0:  
    print("Plese enter a positive integer")  
else:  
    print("Fibonacci sequence is:")  
    for i in range(x):  
        print(recur(i),end=" ")
```

Output:

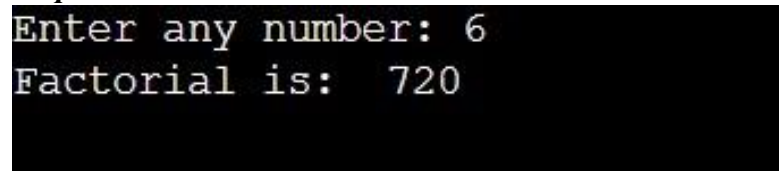
```
Fibonacci sequence is:  
0 1 1 3 5 7 9 11 13 15
```

Programming in Python

Q33. Write a program in Python to get the factorial of non-negative integer using recursion.

```
def fact(n):  
    if n == 0:  
        return 1  
    else:  
        return n*fact(n-1)  
  
n=int(input("Enter any number: "))  
result = fact(n)  
print("Factorial is: ",result)
```

Output:

A screenshot of a terminal window with a black background and white text. It shows the input '6' and the output '720' for the factorial program.

```
Enter any number: 6  
Factorial is: 720
```